DoangJoo "Alan" Synn

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Research Interests

My research builds high-efficiency and scalable machine learning systems, with a focus on automated dataloader tuning, memory-aware and distributed training, and privacy-preserving inference. I work across cloud and distributed computing to make ML training and deployment faster, more resource-efficient, and easier to use.

Education

Georgia Tech, Ph.D. in Computer Science — Aug. 2022-Present
- School of Interactive Computing; advised by Profs. Oh & Ha
- Research: computational design, mechanical artifacts, and motion synthesis (HCI/Graphics)

Korea University, M.Eng. in ECE — Mar. 2020-Feb. 2022
- Thesis: *On Efficient Data Delivery for Machine Learning Systems*

Korea University, B.Eng. in EE — Mar. 2012-Feb. 2020

Experience

Ph.D. Research Intern, Dolby Laboratories — May 2024-Aug. 2025
- Built a time-series, multimodal datastore supporting low-latency media analytics in the Experience Delivery Lab.

MLOps Engineering Intern, LG Uplus Corp. — June-Aug. 2023
- Led **DeepCrunch**, a model-compression library supporting multi-backend AI inference for efficient edge deployment.

Research Scientist, Protopia AI Inc. — Jan. 2021-Aug. 2022
- Built **Stained-Glass**, a privacy-preserving deep-learning framework enabling inference without raw-data exposure.

Cloud Architect · AWS Educate Cloud Ambassador — 2020-2022
- Led engineering for four startup services, translating ambiguous product requirements into production backend systems.

Server Engineer, Flysher Inc. — Jan.-Dec. 2019
- Shipped backend features for four social-game titles; one reached Facebook's global top 10.

System Engineer, AKA Intelligence (Musio) — Apr. 2017-Jan. 2019
- Designed embedded Linux boards and firmware for **Musio**, a deployed social robot.

Selected Publications

1 **DoangJoo Synn**, XinYu Piao, JooYoung Park, and Jong-Kook Kim. "Distributed Training of Deep Learning Models Using Micro Batch Streaming." *KIPS Transactions on Software and Data Engineering*, 2023. doi:10.3745/KTSDE.2023.12.3.101.

2 **DoangJoo Synn** and Jong-Kook Kim. "Num Worker Tuner: An Automated Spawn Parameter Tuner for Multi-Processing DataLoaders." *Proceedings of ACK 2021*, 2021. doi:10.3745/PKIPS.y2021m11a.446.

3 Byung Hoon Ahn, **DoangJoo Synn**, Masih Derkani, Eiman Ebrahimi, and Hadi Esmaeilzadeh. "Protopia AI: Taking on the Missing Link in AI Privacy and Data Protection." *Neural Information Processing Systems Demonstrations*, 2021.